

Kush Ashvinbhai Patel

M.Tech – ICT (Machine Learning)

✉ kushp756@gmail.com

☎ +91 9173149905

🌐 [LinkedIn](#)

🐙 [Github](#)

🌐 [Portfolio](#)



Education

M.Tech (ICT) in Machine Learning, Dhirubhai Ambani University

CPI: 9.39/10

2025 – Present

Gandhinagar, Gujarat

B.E (CE), Madhuben & Bhanubhai Patel Institute of Technology (CVMU)

CPI: 7.81/10

2021 – 2025

Anand, Gujarat

Class 12th, Shree Vasishtha Vidhyalaya (CBSE)

Percentage: 83%

2021

Surat, Gujarat

Class 10th, Shree Vasishtha Vidhyalaya (CBSE)

Percentage: 88%

2019

Surat, Gujarat

Experience

Dhirubhai Ambani University

Teaching Assistant – Data Structures & Algorithms (C++)

Aug 2025 – Present

Gandhinagar, Gujarat

- Mentor and guide undergraduate students through complex **DSA concepts in C++**, including trees, graphs, dynamic programming, and sorting, strengthening problem-solving foundations for 30+ students.
- Design weekly **algorithmic challenges** and rigorously evaluate code submissions for correctness, **time & space complexity**, preparing students for competitive technical interviews.
- Conduct doubt-clearing sessions and provide one-on-one feedback, resulting in measurably improved student performance and course engagement.

HP Param IT Solutions

Flutter Developer Intern

Jan 2025 – May 2025

Remote

- Built and deployed a **cross-platform mobile application** using **Flutter** for an IoT-based *Street Light Automation* system, targeting Android and iOS from a single codebase.
- Collaborated with backend engineers to integrate **RESTful APIs**, enabling live device-status updates and remote actuation with seamless real-time hardware control.

Projects

Beyond Visible Spectrum: AI for Agriculture (ICPR 2026) | PyTorch, ViT, EfficientNet, ResNet [[github](#) – [kaggle](#)]

- Achieved **3rd Place globally in Task 2** (Synthetic Hyperspectral Data Generation) out of **574 participants across 328 teams** in the ICPR 2026 Beyond Visible Spectrum competition, winning a **£40 Amazon prize**.
- **Task 1** (Crop Disease Classification): ranked **Top 25** using a **multi-model ensemble** of ResNet, EfficientNet, and ViT on hyperspectral & multispectral remote sensing imagery with spectral band preprocessing.
- **Task 2** (Synthetic Data Generation): developed a self-supervised approach to generate **synthetic Sentinel satellite data**, boosting crop disease classification accuracy on scarce real-world agricultural field imagery.

AQI Vision Predictor | PyTorch, EfficientNet, CNN, OpenCV, Python [[github](#)]

- Built a **computer vision pipeline** to predict **Air Quality Index (AQI)** directly from real-world images and camera input, reframing a traditionally sensor-dependent problem as an image classification task.
- Fine-tuned **EfficientNet** as the primary backbone alongside custom **CNN architectures**, applying transfer learning and data augmentation to improve generalisation across diverse environmental conditions.
- Designed end-to-end preprocessing with **OpenCV** (haze estimation, colour channel analysis) to extract visually discriminative features correlated with particulate matter and pollution levels.

Earth Observation: Delhi Airshed Land-Use Classification | PyTorch, ResNet18, GeoPandas, Rasterio [\[github\]](#)

- Built a **land-use classification pipeline** for the Delhi-NCR airshed using **Sentinel-2 multispectral satellite imagery**; performed spatial joins (GeoPandas, Rasterio) to filter **8,015 valid patches** from 9,216 total across the NCR boundary.
- Automated **ground-truth label construction** by cross-referencing each patch against the **ESA WorldCover 2021** raster (10m resolution), mapping 11 ESA classes into 5 simplified land-use categories via majority-vote sampling.
- Fine-tuned **ResNet18** (ImageNet pretrained) for 5-class prediction using stratified train/val/test splits, evaluating with accuracy, macro F1-score, and a full confusion matrix on held-out test data.

Sleep Apnea Detection from Physiological Signals | PyTorch, 1D CNN, SciPy, Python [\[github\]](#)

- Developed an automated **sleep apnea detection system** using overnight physiological signals (nasal airflow, thoracic movement, SpO₂) from **5 participants**, processing **8,800 labelled 30-second windows** with 50% overlap.
- Applied **Butterworth bandpass filtering** (0.17–0.4 Hz) for breathing-rate isolation and trained a **3-layer 1D CNN** with inverse-frequency class weighting to handle severe class imbalance (91% Normal vs. 7% Hypopnea vs. 2% Apnea).
- Evaluated using **Leave-One-Participant-Out (LOPO) cross-validation** to test generalisation across unseen individuals, achieving **67% apnea recall** – prioritising detection of the most clinically critical events.

Technical Skills

Languages: Python, C/C++, SQL, Dart, JavaScript

ML / DL Frameworks: PyTorch, Scikit-learn, TensorFlow, OpenCV, Pandas, NumPy, HuggingFace Transformers

Web & Mobile: Flutter, Flask, FastAPI, React

Developer Tools: Git, Docker, Postman, Jupyter Notebooks, Linux Shell, VS Code

Cloud & Databases: AWS (EC2, Fargate), MongoDB, MySQL

Core CS: Data Structures & Algorithms, Operating Systems, DBMS, OOP, Computer Networks

Position of Responsibility

- Core Member – Technical Wing, **AI Club, Dhirubhai Ambani University** (Aug 2025 – Present), organising workshops on Deep Learning, Computer Vision, and LLM-based agentic systems for **100+ club members**.
- Teaching Assistant – **Data Structures & Algorithms (C++)**, Dhirubhai Ambani University, mentoring 30+ undergraduate students through weekly sessions, lab evaluations, and one-on-one doubt clearing.

Achievements

- [ICPR 2026 – Beyond Visible Spectrum \(3rd Place\)](#): Ranked **3rd globally in Task 2** out of **574 participants / 328 teams**, winning a **£40 Amazon prize** for synthetic Sentinel satellite data generation for crop disease classification.
- [Kaggle – Bidding Predictions for Construction \(Solo #1 / #2\)](#): Ranked **1st on Public** and **2nd on Private Leaderboard** (solo) using a multi-stage **LightGBM + XGBoost + CatBoost** ensemble with Bayesian-smoothed hierarchical pricing on 832K rows. [\[github\]](#)
- **GATE Qualified (Dual)**: Cleared GATE in **Computer Science** (Score: 456) and **Data Science & AI** (Score: 413) demonstrating strong command of core CS and AI fundamentals.
- **Competitive Programming**: [LeetCode](#) – Max Rating **1682**, 100+ solved; [CodeChef](#) – **3-Star**, Max Rating **1658**, 180+ solved; [Codeforces](#) – Max Rating **1089**, 50+ solved.
- **AI & ML Certifications**: 5-Day Gen AI Intensive (Google & Kaggle); Machine Learning (IBM); TensorFlow Developer; Deep Learning & ML Specializations (DeepLearning.AI).